

than 117,000 packages for different functionalities and applications in PyPi. This escalated to 123,086 packages by November 30, 2017.

The table in Figure 1 gives the statistics fetched from *ModuleCounts.com*, which maintains data about modules, plugins and software tools.

Date	Nov-24	Nov-25	Nov-26	Nov-27	Nov-28	Nov-29	Nov-30
Packages in PyPi	122,619	122,669	122,723	122,808	122,918	123,008	123,086

Figure 1: Statistics of modules and packages in PyPi in the last week of November
(Source: <http://www.modulecounts.com/>)

Python based packages for secured backup and recovery

As Python has assorted tools and packages for diversified applications, security and backup tools with tremendous functionalities are also integrated in PyPi. Descriptions of Python based key tools that offer security and integrity during backup follow.

Rotate-Backups

Rotate-Backups is a simplified command line tool that is used for backup rotation. It has multiple features including flexible rotations on particular timestamps and schedules.

The installation process is quite simple. Give the following command:

```
$ pip install rotate-backups
```

The usage is as follows (the table at the bottom of this page lists the options):

```
$ rotate-backups [Options]
```

The rotation approach in Rotate-Backups can be customised as strict rotation (enforcement of the time window) or relaxed rotation (no enforcement of time windows).

After installation, there are two files *~/rotate-backups.ini* and */etc/rotate-backups.ini* which are used by default. This default setting can be changed using the command line option *--config*.

The timeline and schedules of the backup can be specified on the configuration file as follows:

```
# /etc/rotate-backups.ini:
[/backups/mylaptop]
hourly = 24
daily = 7
weekly = 4
monthly = 12
yearly = always
ionice = idle
[/backups/myserver]
daily = 7 * 2
weekly = 4 * 2
monthly = 12 * 4
yearly = always
ionice = idle
[/backups/myregion]
daily = 7
weekly = 4
monthly = 2
ionice = idle
[/backups/myxmc]
daily = 7
weekly = 4
monthly = 2
```

Option	Description
-M, --minutely=COUNT	Number of backups per minute
-H, --hourly=COUNT	Number of hourly backups
-d, --daily=COUNT	Number of daily backups
-w, --weekly=COUNT	Number of weekly backups
-m, --monthly=COUNT	Number of monthly backups
-y, --yearly=COUNT	Number of yearly backups
-I, --include=PATTERN	Matching the shell patterns
-x, --exclude=PATTERN	No process of backups that match the shell pattern
-j, --parallel	One backup at a time, no parallel backup
-p, --prefer-recent	Ordering or preferences
-r, --relaxed	Strict rotation with the time window for each rotation scheme
-i, --ionice=CLASS	Input-output scheduling and priorities
-c, --config=PATH	Configuration path
-u, --use-sudo	Enabling the use of 'sudo'
-n, --dry-run	No changes, display the output
-v, --verbose	Increase logging verbosity
-q, --quiet	Decrease logging verbosity
-h, --help	Messages and documentation